

CHAPTER 1

TECHNIQUE



PHOTO: BJØRN HELGE RØNNING

The lack of good holds for hands and feet emphasises the importance of good technique. Siri Olimb Myhre gliding up Oppvarmingseggen (Font 6c+), Harbak, Norway.

CLIMBING IS DEFINED as a technical strength sport. In other words, technique is one of the most important elements to master, but for many it will also be the most difficult. So, what is technique, and, perhaps more importantly, what is good technique? If you see a child who has never moved up a wall before, it will have an intuitive understanding of maintaining balance on the wall. This is an example of good technique. Similarly, we often see that beginner climbers who have a good physical foundation tend to rely on their arms to get up the wall. This is an example of poor technique. Technique therefore has to do with maintaining balance and conserving power. But is this all, and is there a universal answer to what is good technique? We want to argue that there are a number of basic movements and positions that are universal. At the same time, physical and mental abilities and limitations will allow for great variation in how the various techniques are used.

This chapter presents grip positions, footwork, foundations and specific techniques. We also look at which of these are suitable for different types of climbing. Finally, there are tips on how to train technique.



PHOTO: BJØRN HELGE RØNNING

Stian Christophersen om Fokus (Font 8a+), Harbak, Norway. Stian is crimping hard with his left hand while reaching for the crux pinch.

GRIP POSITIONS

It's important for us as climbers to master the different types of grip positions. Holds, like slopers, pockets and crimps – and how best to grab them – vary greatly. Many routes and boulder problems demand use of several, if not all, of the different grip positions. We have chosen to categorise grip positions into the following types:

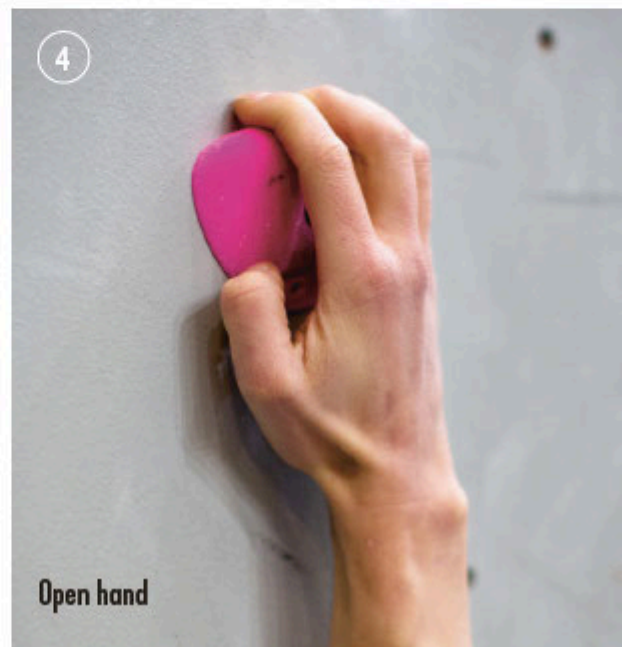
- open hand

- pinch

- half crimp

- crimp

Some grip positions are more aggressive than others. An aggressive grip position means that the middle joint in the fingers is bent at an angle of 90 degrees or more, which leads to a greater activation of the shoulders and arms. An aggressive grip position allows for bigger and more explosive movements, while a less aggressive grip lends itself to more careful and precise climbing. The grip position is often dictated by the hold, but the use of different types of grip position on the same hold can also allow for different ways of moving your body.



Holds 1–3 are what we call pockets.

OPEN HAND

This is the most common grip position and it can be used on most holds. Most of the flexion and load is focused on the outermost joint. Climbers can use one to four fingers in this position. The number of fingers engaged is usually based on the shape and size of the hold.

It's worth noting that once you add the pinkie finger to this grip position, you engage your arm and shoulders and the grip position becomes more aggressive. This is because when you engage the pinkie the angle of the middle joint on the other three fingers is reduced.



This photo shows the correct grip position on a sloper, providing a large contact area between the hold and the hand, and hence more friction.



This photo shows a half-crimp grip position on a sloper, which reduces the contact area and therefore the friction.

The open-hand grip position is especially handy for large holds and pockets, but can also be used for smaller edges as long as they aren't too thin. Open hand grip is the preferred grip position for lead climbing as the physical cost is lower than with other types of grip position. It is assumed that the fingers are less prone to injuries when using an open-hand grip position instead of more aggressive grip positions.

When climbing on slopers, as in holds that have a negative angle, a variant of the open-hand grip position is usually used. The goal is to maximise the contact area between the hold and the hand and fingers. More contact area means more friction, as demonstrated in the photos.



PINCH

As the name implies, this grip position is for when you have to pinch around both sides of a hold. Your thumb is on one side while your other fingers are countering on the other side. By squeezing the hold like this you can keep body tension on the wall in ways which other grip positions won't allow. But pinching requires strength and many climbers struggle with this grip position.

On indoor climbing walls the holds are mounted in such a way that they protrude from the wall and this allows climbers to pinch almost any indoor hold. Some types of rock, like limestone with tufa features, also allow for pinching to be used outside.



HALF CRIMP

The half crimp is suitable for smaller holds and edges. The middle joint of the fingers is bent, and the outer joint is straight. The half crimp is a more aggressive grip position than the open hand and is better suited for when the moves are long or the wall is steep and the positions demanding. Half crimp is commonly used in bouldering and for crux sections when lead climbing. Using a half crimp is more physically taxing than climbing with an open hand because more muscles are involved, but it also allows for a higher exertion of force.



CRIMP

Crimping is suited to really small edges and when moving explosively between small holds. The grip position is similar to the half crimp, but you also engage the thumb to lock the fingers in a bent position. This is the most aggressive grip position and it should only be used when necessary.



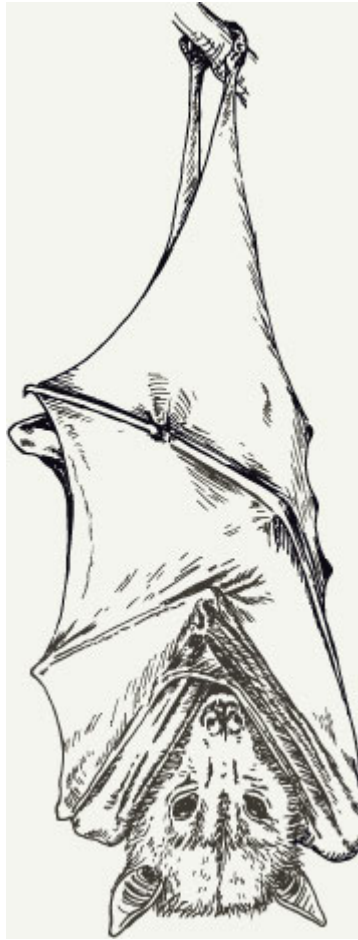
Children and adolescents whose finger joints are still developing should not crimp.

It might seem that this should be the preferred grip position for any situation, and for many climbers it is. The catch is that crimping is the most aggressive type of grip position, and it is closely related to injuries to pulleys, cartilage, bones and sinews. Crimping with care and learning how and when to use the other types of grip positions can reduce the chance of injuries, and as the crimp has some clear performance advantages we are going to cover the topic and keep on crimping.

WHY DO WE CRIMP?

Have you ever wondered how bats manage to hang upside down when they sleep? And what does this have to do with crimping?

As many of you have probably experienced, it's easier to hold on to smaller holds when you lock your thumb over the outermost part of the index finger than if you use an open hand or half crimp. Many also resort to crimping when they are getting pumped, so it's obvious that this grip position provides some advantages. But why?



There are several reasons why we might choose to crimp small holds. First and foremost, we get help from an extra finger, the thumb, to generate force in the crimp. By crimping instead of using an open-hand grip position, we also increase the contact area between the hold and the fingertips, providing better friction and allowing us to hang on to smaller holds and in-cuts. These two factors can explain why we choose to crimp, but there is more. When crimping, the angle of the finger joints is in a better position for exerting power than when using an open hand. In addition, the friction between the tendon and the tendon sheath will increase, and this friction is estimated to account for up to nine per cent of the force exerted at the middle joint of the finger. This means that we can either relieve the musculature by nine per cent or get nine per cent more power in a grip position at maximum muscular exertion. This mechanism is called the Tendon Locking Mechanism (TLM) and is the main reason why bats can hang by their toes and sleep with the minimal use of muscular forces.

VARYING GRIPS

Many climbers favour a particular type of grip position and this often has to do with genetic factors. Climbers with small hands prefer edges and crimps and will be at a disadvantage on wide pinches where large hands are an advantage. It is important to train and practise with all the different grip positions in order to be able to adapt to the various challenges you'll face on the wall. That being said, many climbers are also able to use their favourite grip position on most holds. Climbers who spend a lot of their time outdoors often resort to crimping, even when they venture indoors. This is not necessarily wrong, but anyone who wants to become a complete climber should clearly train and practise all the different grip positions. Because of the risk of injury, children and beginners should avoid crimping.



FOOTWORK

Your legs and feet can and should take a large part of the force needed to keep you on the wall. This requires proper positioning of the feet. There is a big difference between how you should place your feet when smearing your shoes on a sloper or edging your toes on a small edge, and walls at different angles add to the complexity. There are also different shoes designed for different applications.

First and foremost, it is important to place your foot precisely on the foothold. It is important that you as a climber look at your foot as you place it. Too often, we see climbers falling because their foot slipped after they placed it haphazardly without looking. Particularly for beginners, this is a recurring problem. It is easy to forget about your feet because you are looking up and focusing on the next handhold. And when nerves or stamina starts to become an issue, imprecise footwork is common. Remember that a small foothold can have a tiny recess that's hard to spot, which can be crucial for whether your foot will stick or slip. In other words, footwork requires precision and focus. Take a moment to observe experienced climbers on the wall; it is often impossible to hear when they place their foot on a hold. Their footwork is extremely precise so that they can get the most out of their feet.



- Look at the foothold and place your foot gently and with precision.**
- Lower both your body and your heel when smearing on larger holds.**



It's important to lower both your body and your heel when smearing on volumes or larger holds in order to increase the contact area, and hence the friction, between the hold and your shoe.



Both the inside and the outside edge of the shoe, as well as the tip, can be used for edging on small holds.

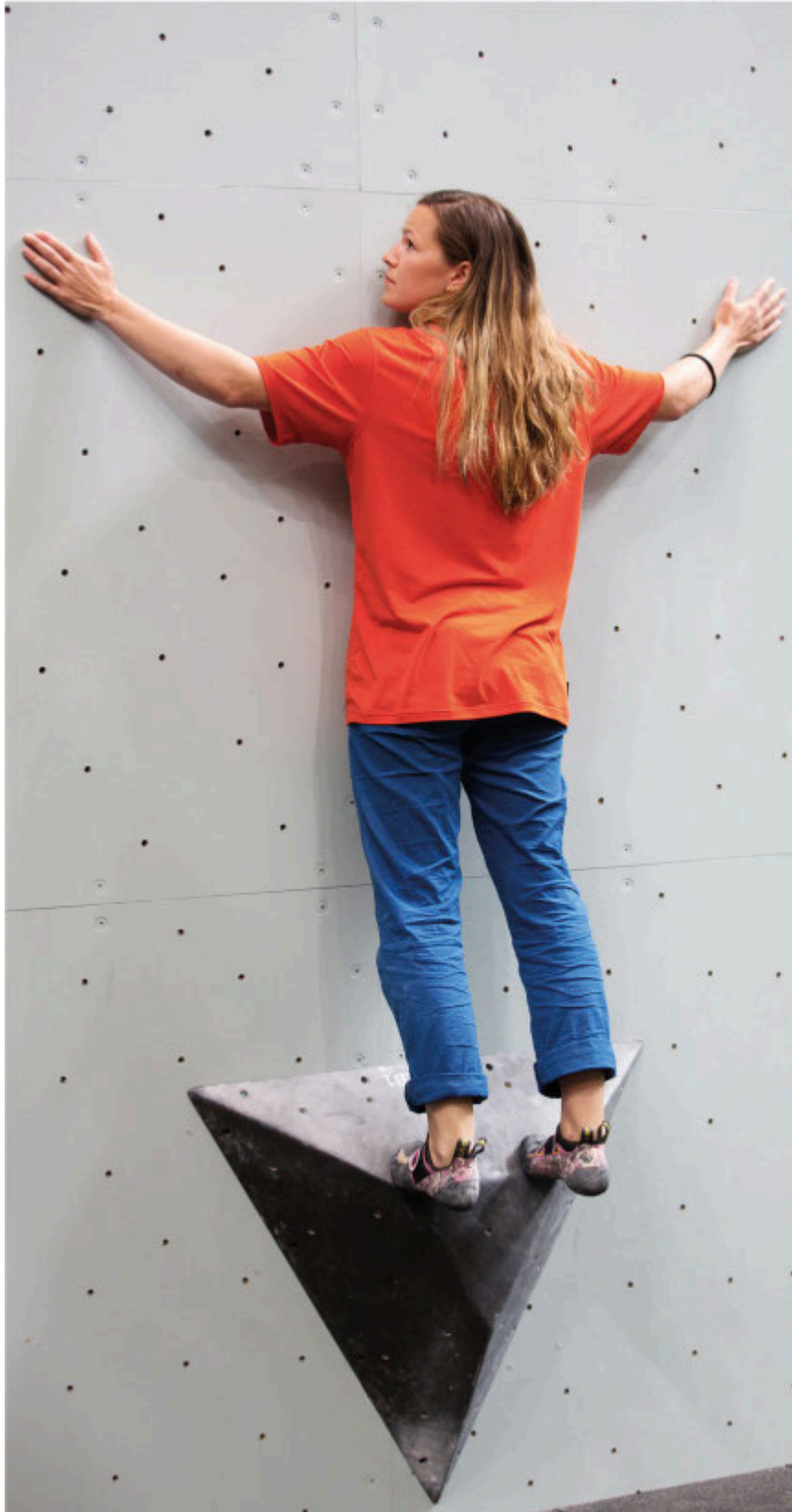
1. EDGING

On small edges it is important to use the edging properties of your climbing shoes. Climbing shoes are relatively stiff, and by precisely placing the edge of the shoe you can quite securely stand on holds that are just a few millimetres deep. You can edge with the outside edge, the inside edge and the tip of the shoes. Which technique is best depends on your body position and the foothold. If you're standing sideways against the wall, use the outside edge or the tip of your shoe. If you are facing the wall, use the inside edge or the tip of your shoe. On very small footholds it will be best to use the tip of your shoe as this is what most shoes are designed for. Not all shoes are created equal, and which model and brand is best for which purpose will vary, but as long as the shoes aren't too soft, most shoes will provide good support on small footholds. But be aware that old and worn shoes will have lost much of their edging properties, so if you know you're going to be climbing on the edge you should think about which shoes to use.



2. SMEARING

When standing on slopers, volumes or directly on the wall, it is the friction between your shoe and the foothold that enables you to stand. This is known as smearing. Imagine trying to rub or smear the rubber of your shoes on the foothold. The contact area between your shoe and the foothold is crucial for how well your foot will be able to stick to the foothold. Lowering your heel will increase the shoe's surface area in contact with the foothold. If you try to use the edge of your climbing shoe, the contact area will be smaller which will increase the risk of your shoe losing grip and slipping off. You can smear on basically any surface, and the technique is often used to maintain balance and as an intermediate step when moving your feet up to better footholds.



3. VOLUMES AND LARGE FOOTHOLDS

When standing on a large foothold that protrudes significantly from the wall, it is advantageous to stand as far out on the foothold as possible. The reason for this is that you can lean more towards the wall and the load on the handholds is therefore lower. Very often we see inexperienced climbers place their feet closer to the wall when standing on volumes because this feels safer. This greatly increases the load on the fingers and arms as, in practice, you're making the wall steeper.



CLIMBING SHOES

The steepness of the wall will often dictate how you use your feet. If the angle of the wall is less than 90 degrees (a slab), it is generally better to use soft shoes so that you get the full effect of the rubber smearing on the wall. On vertical walls, stiffer shoes with good edging qualities are more appropriate. On steep overhangs, aggressive shoes will allow you to pull on the footholds with your toes. Aggressive shoes have a downturned, banana-like shape that helps translate force from your foot on to the foothold.

Indoors, and on most occasions outdoors, it's necessary to have shoes that offer both edging and smearing capabilities. Many shoe models combine these qualities well, and they are well suited for beginners and indoor climbers. As you gain more experience, it can be wise to carry several different models in your bag for different applications. An aggressive shoe for steep climbing, a soft shoe for smearing indoors and a rigid shoe for outdoor climbing.



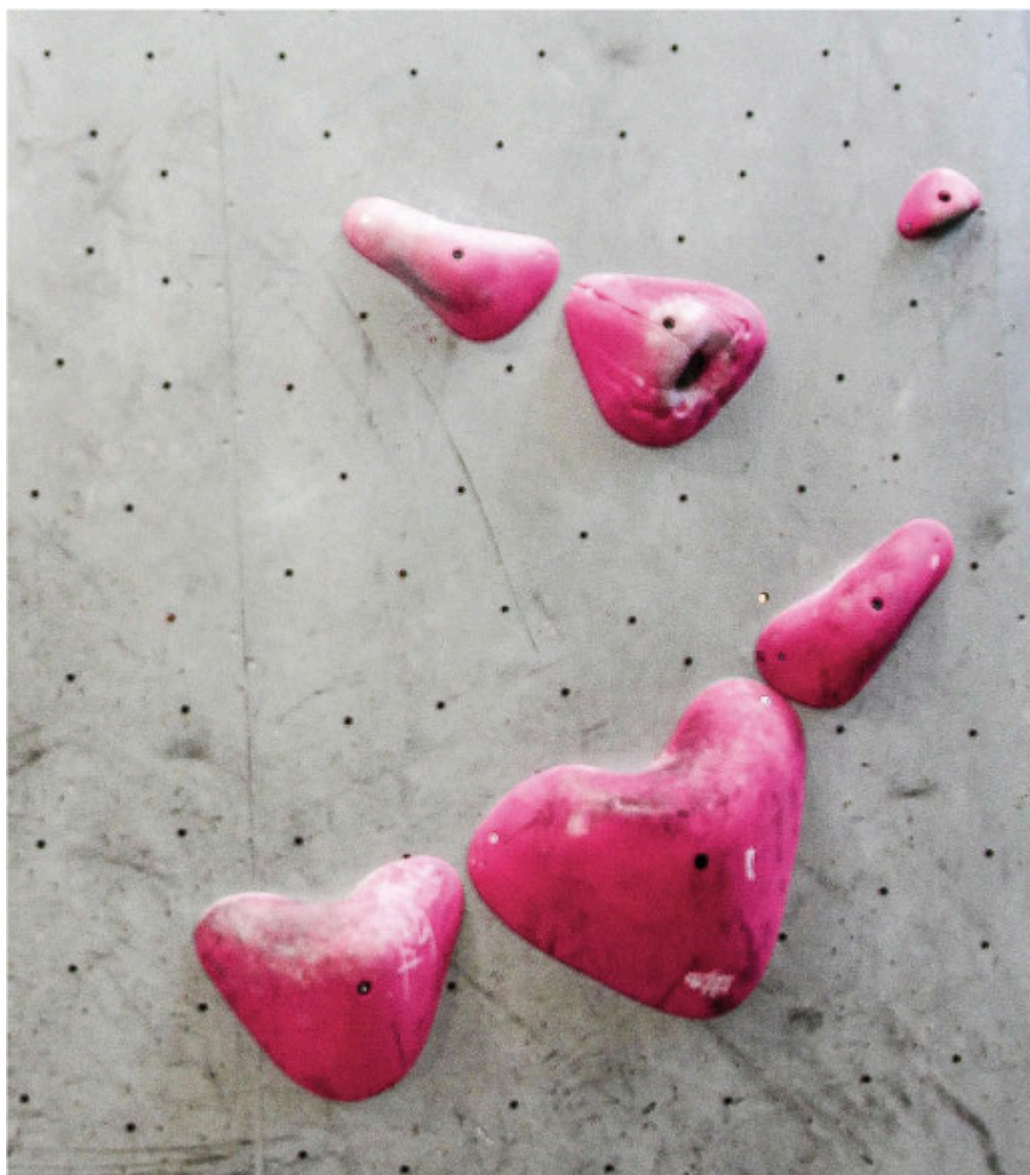
PHOTO: VOLKER SCHÖFFL, REPRINTED WITH PERMISSION

X-ray image of a foot wearing a climbing shoe. Notice how the toes are squeezed together by the tight-fitting shoe. Find a shoe that fits your foot, and let your feet rest by regularly taking your shoes off during your training or climbing session.



BUYING SHOES

The number one factor when buying shoes is the fit! The fit of climbing shoes varies wildly between different brands and models. Be the annoying customer in the shop and try on all the different models they stock in order to make sure you find the shoe that best fits your foot. Climbing shoes will adapt to your foot and expand somewhat with use, but it is important that your foot fills the entire volume of the shoe, both in the toe box and in the heel cup. It's also important that the shoes aren't too small – too often beginners are advised to get the smallest shoes they can fit their feet into, and that pain is to be expected. The point of tight-fitting shoes is to maintain a stiff front tip, which is only really necessary when climbing outdoors on really tiny footholds. The smearing capabilities are also reduced if the shoe is too small.



THE FUNDAMENTALS

The fundamentals are the founding principles to which we must always adhere, regardless of what type of movement we are about to execute. We have chosen to define the following fundamentals:

- balance and weight distribution
- force vector and tension
- static and dynamic climbing styles.

Our reason for choosing these three is that they are essential for any movement we perform on the wall. No matter how we move, there's a redistribution of weight, and there's always a choice between doing a move with speed (dynamically) or by locking off (static). In order to use a hold in the best possible way, we're dependent on the force acting in the correct direction. Often we also utilise the force acting between two points of contact on the wall, for example between two handholds or between a handhold and a foothold. This is what we call tension.



The climber lowers her body underneath the handhold in order to relieve the arms and achieve a better angle relative to the handhold.



The centre of gravity is centred between a relatively wide area of support.

1. BALANCE AND WEIGHT DISTRIBUTION

It's important to keep your balance in order to expend as little energy as possible. In sports science we define the athlete to be in balance when the centre of gravity is centred on the area of support. The centre of gravity is the point where the weight of your body is centred, usually just below the navel. The area of support is the area between the points of your body touching the ground. If you're standing on the ground and place your legs on each side of your body you'll be standing safe and securely because your centre of gravity is centred on your area of support. If, on the other hand, your centre of gravity is outside your area of support, you'll be off balance. A wider area of support will give you more room to move without losing your balance. The traction of your points of contact will also increase the amount of movement you can perform securely without losing balance: for example, it's easier to maintain your balance during challenging movements when standing on concrete compared to when standing on ice.

As with any other sport, the climber is in balance when the centre of gravity is confined within the area of support. The area of support for a climber is defined by the contact points on the climbing wall. This includes not just your two feet, but also your hands. A stable balance is best achieved by centring the centre of gravity on the area of support. In order to maintain balance try to always have four points of contact (two hands and two feet) on the wall before initiating a move, and three points when executing a move. In effect, this means that if you want to move one hand, you should have a secure grip position with your other hand and both your feet. A better hold will also increase your ability to move freely on the wall. When climbing on large and positive holds you can lean out from the wall and from side to side. This means you can compensate for poor balance by using better holds, but this isn't possible when using smaller and less positive holds.



In order to maintain good balance, the climber moves her body to the right before she moves her right arm.

As a general rule you should lean in to the wall and below the handholds as much as the available footholds allow. By lowering your body underneath the handholds you will lessen the load on your arms and your body will have a better angle in relation to the handholds. In order to avoid getting more fatigued than is necessary, it is best to hang with your arms straight whenever possible.

Weight transfer is the principle of positioning the body, especially the hips, so that the centre of gravity remains centred on the area of support as we move. The most common form of weight transfer is to move the centre of gravity to the right when moving the right hand, and vice versa. Maintaining your balance usually doesn't require large movements. Quite often, just shifting your hips to the side by a few centimetres is all it takes to stay in balance. By proficient use of weight transfer, you will achieve better balance when moving, increase reach and waste less energy. Weight transfer is the principle behind a large proportion of the techniques we use when climbing and is a basic skill to practise.



Remember that you can place a foot anywhere on the wall to increase the area of support and achieve better balance.



It is important to remember that you must position your body so that it is in balance for the position it will be in after you have made a move. Your body may well be in balance as you start a move, but this is of little help if your body is unbalanced when the move is completed. The clearest example of this is the ‘barn door’ – when the climber loses balance and swings out from the wall.

SIDEPULL EXERCISE

The exercise consists of one right-facing sidepull and three footholds. The three footholds should be evenly spaced horizontally: one foothold directly below the sidepull, and one foothold on each side. Hold the sidepull with your right hand and use only your right foot to stand on the wall. Do not flag with your left foot or use other tricks. It will be easy to maintain balance if you stand on the foothold to the right. You’ll also be able to maintain balance when standing on the middle foothold. But you’ll lose balance if you stand on the foothold to the left of the sidepull.



Hannah Midtbø using opposing forces between holds to climb a steep boulder problem.

2. DIRECTION OF FORCE AND TENSION

It is only possible to use many of the holds we use when we're climbing because we can apply force in the right direction. Consider a sidepull. If we try to use it like a normal hold and pull straight down, we can't hang on. Instead we have to try to position ourselves so that the force is acting perpendicular to the grip surface. In the same way we have to step up high with our feet in order to use an undercling. Only when we get enough force acting up into the hold are we able to effectively use the undercling.

Sometimes we can use the force acting between holds. We call this tension, which is a term covering all situations where the body is maintaining tension or forces between different points of contact. This can be between two handholds, but also between a handhold and a foothold.

Picture yourself squeezing together two opposing sidepulls in order to achieve tension. We call this compressing between holds, or compression climbing, and this is particularly important when climbing on slopers.

Tension gives us many more options for climbing on holds that wouldn't normally be good enough to hang.



Synnøve Berg Nesheim using a toe hook to create enough tension on a left-hand sidepull.



When climbing overhangs you need to maintain tension all the way through your body, from your fingers to your toes. Understanding how to optimise this tension is difficult. You have to pull with your arms – but not too much

or you'll lose contact with your feet! To understand this balance it can be helpful to think of your feet as claws that pull down at the same time as your arms are pulling up.



Your body is stronger than your fingers! If you are exerting more force with your body than is necessary to maintain your position, you might unnecessarily increase the load on your fingers. Optimal load balance requires experience and a lot of time on the wall, and emphasises the importance of practising as much as possible on all kinds of holds and angles.

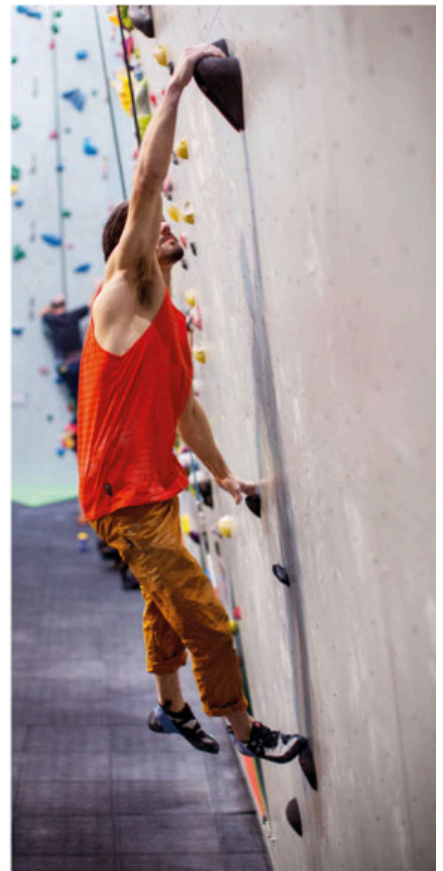


A static movement pattern is the style of choice when the moves need to be precise and the holds are hard to hit.

3. STATIC AND DYNAMIC CLIMBING STYLE

One of our old climbing coaches and a legend of competition climbing in Norway, Chris Fossli, once said that all movements should optimally be dynamic, but that our ‘poor’ dynamic abilities inhibit them from being so. This is probably at least partly correct, and we should strive to perform most moves with at least some dynamic elements in order to minimise energy waste. Despite this, where a static style might require a certain level of strength in order to make a move slowly, faster and more dynamic moves will require a faster rate of force development, coordination of arms, legs and body, spatial awareness, and contact strength.

A static move is where one arm is held statically in place without changing its angle, while the other arm moves to the next hold. By definition, all moves are dynamic, but in climbing we use the term dynamic to refer to moves that are quick, and where both arms are in motion. In the same way we use the terms static and dynamic climbing style to describe the way different climbers tend to move on the wall. Static climbers like to move slowly and controlled, while dynamic climbers move faster and ‘bounce’ between holds. Static climbing is useful when the holds are small or hard to hit, or when climbing on sight when you don’t know what and where the next hold is. You reduce the chance of missing the hold, but often end up using more power to lock off. A static style is often preferred on vertical and less steep climbing. In these cases, it’s often the size of the holds and the balance required that makes the climbing challenging, and this is more easily overcome with a static and controlled style. Static climbing requires more body strength because you must lock your body in different positions, and climbers with a strong core and shoulders are better suited to this style of climbing.



Big moves are usually performed dynamically.

By executing a move dynamically we can exploit the speed generated at the beginning of a move, without stopping and using unnecessary energy. This is especially relevant for steep climbing. In these cases it is usually the longer moves that are the most challenging, and maintaining the elusive balance is perhaps less crucial. What's more, climbing statically through steep terrain is a lot more physically challenging as it requires more energy to maintain a stable body position during moves.

Climbing dynamically can also be quite effective on slopers. If you pull yourself up slowly on a sloper, your centre of gravity will gradually move away from the wall and the friction will worsen. By dynamically making a move between slopers, you can initiate the move from below, where the angle in relation to the hold is more positive and the friction is better. The principle is the same with poor and slopey footholds. If you pull up, your heels lift, the contact area decreases, and you risk slipping off the foothold. When moving from slopey footholds you should therefore move dynamically and initiate the move from below.

A quick and dynamic move is recognised by a starting phase, a floating phase and a stopping phase, similar to when performing a high jump, or shooting a bullet into the sky. In the starting phase you want to create force using the big muscle groups in your legs and core, but the arms also help to initiate the movement, and guide you in the right direction. It's important that your arms pull your body towards the wall, so that when you grab the next hold your body isn't already moving away from the wall. In the floating phase the arm and the hand must be coordinated in order to precisely hit the next hold.



Deadpointing slopers: the movement starts from below; the centre of gravity is then moved over the left leg and in towards the wall. This allows the left leg to straighten and create an upwards motion, while the right hand maintains grip and keeps the centre of gravity close to the wall. With enough power generated early, more strenuous body positions are bypassed and the next hold is reached at the apex of the move.

A dynamic move obviously needs to be stopped when the next hold is reached. By adapting the amount and direction of power in the starting phase, less energy is needed to stop the movement. A bullet fired straight up into the air will, in principle, be weightless at the apex of its trajectory before it starts to fall back down again. We call

this the deadpoint of the trajectory, and it is the preferred time to stop a move. In any case the movement has to be stopped, which requires contact strength. Contact strength is the ability to quickly generate enough power to maintain grip on the next hold and stop the move. Sticking a dynamic move might also require tension in order to lock the body position. Especially if the handholds are poor.

Many climbers find dynamic climbing to be mentally challenging. It's difficult to truly let go and go all-in for the next hold. It feels much safer to make a move statically and with control. This is especially common in the world of sport climbing because of the fear of falling. Dynamic climbing is something that can and should be practised. Especially for shorter climbers, dynamic climbing can be essential, as some moves can be quite reachy. Remember, getting used to a more dynamic style can take time. We have worked with many climbers who are uncomfortable with the risk involved in doing a move more dynamically, and therefore prefer a more static and controlled style. Yet we find these athletes raise their level significantly when adopting a more dynamic climbing style.

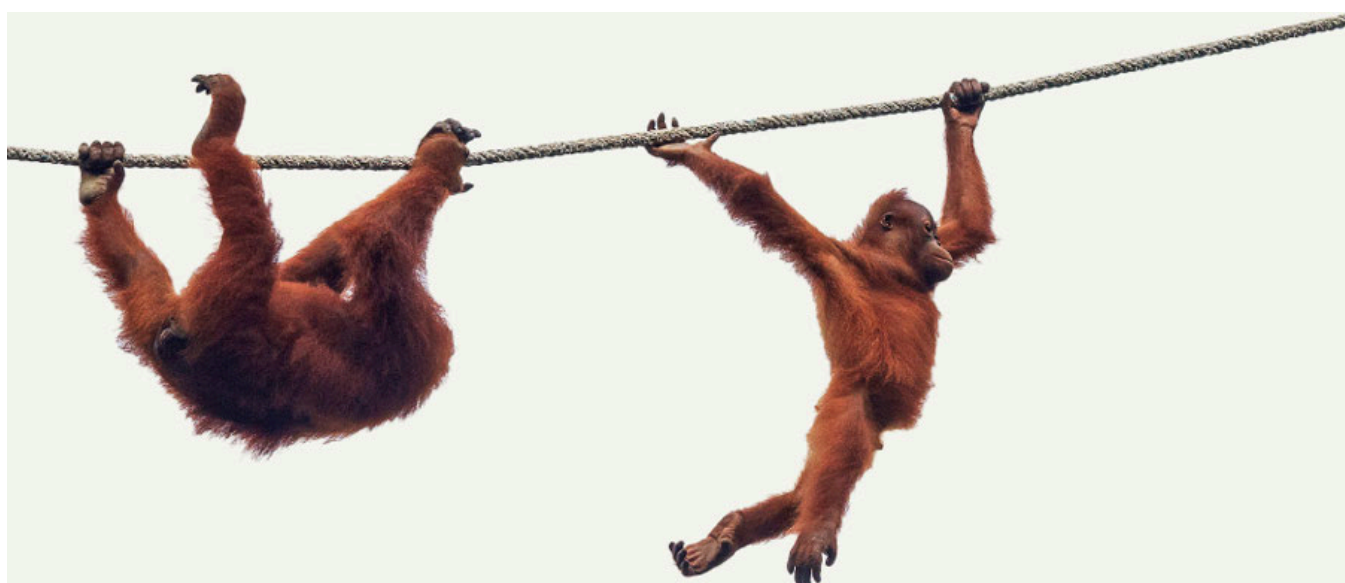


PHOTO: GETTY IMAGES

GERMAN DYNAMICS

When climbing dynamically, the movement tends to flow from one hold to another. Udo Neumann, a former coach of the German national team, likens dynamic climbing to the way a monkey swings from branch to branch. They use the transition between the swings in a perfect way when each pendulum movement flows into the next without visible interruptions. In the same way you as a climber can use the energy generated from a move to keep the momentum going into the next move. Most of us have probably experienced stopping and losing all momentum after a dynamic move, and then having to start again from scratch to generate speed for the next big move, but we often see skilled climbers carrying the speed from one move into the next. There are of course many complicating factors, one being that vines aren't climbing holds, but the principle is the same – use the least amount of energy to execute a move! This means dynamic moves and fluid transitions between each move. 'The perfect combination between fast and slow, aka flow.' – T. Landman.

SPECIFIC TECHNIQUES

In this section we present the specific techniques which are based on the fundamentals presented in the last section. We as climbers strive to choose the different techniques that – based on the challenges we're currently facing on the wall – allow us to use the least amount of energy as possible.

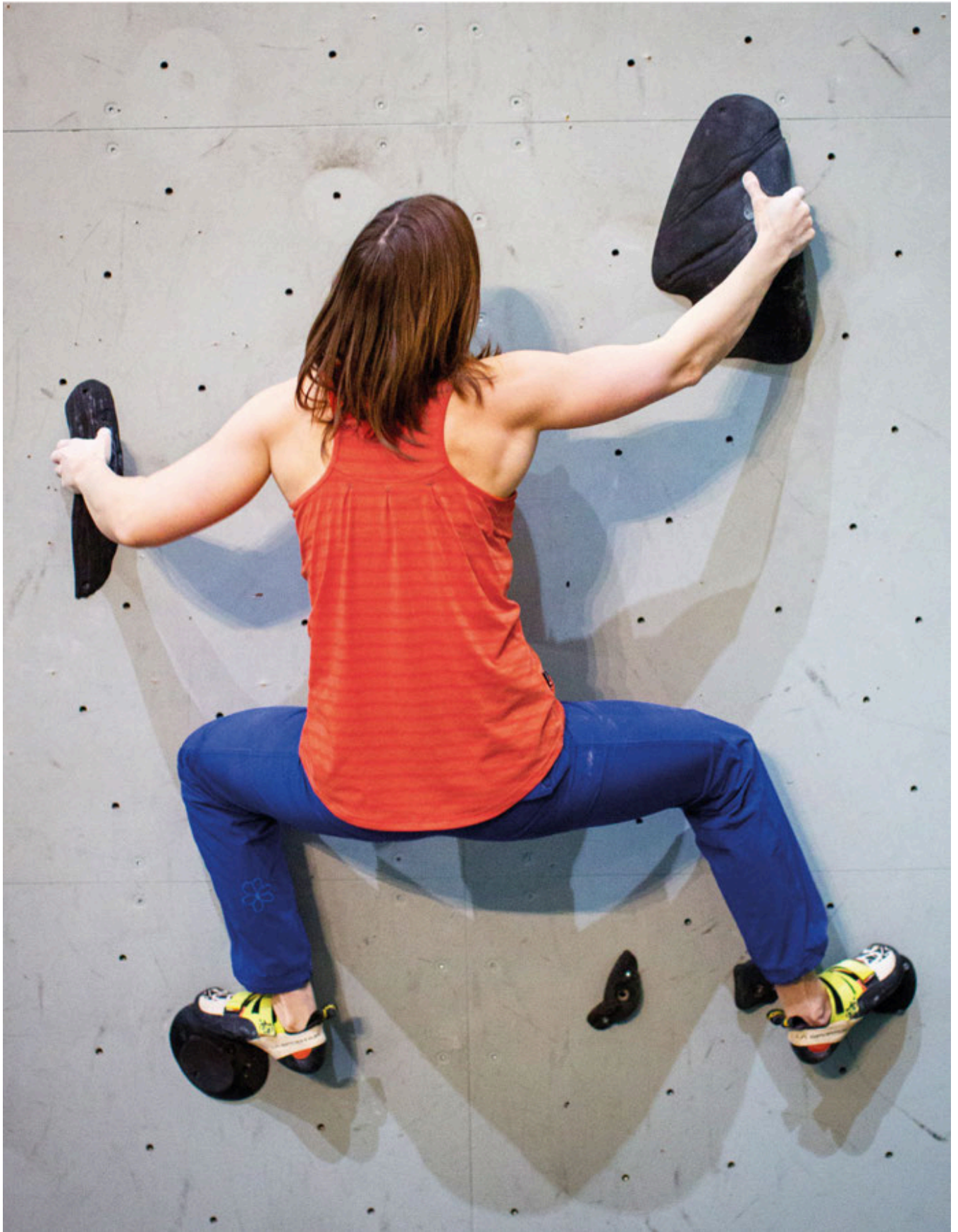


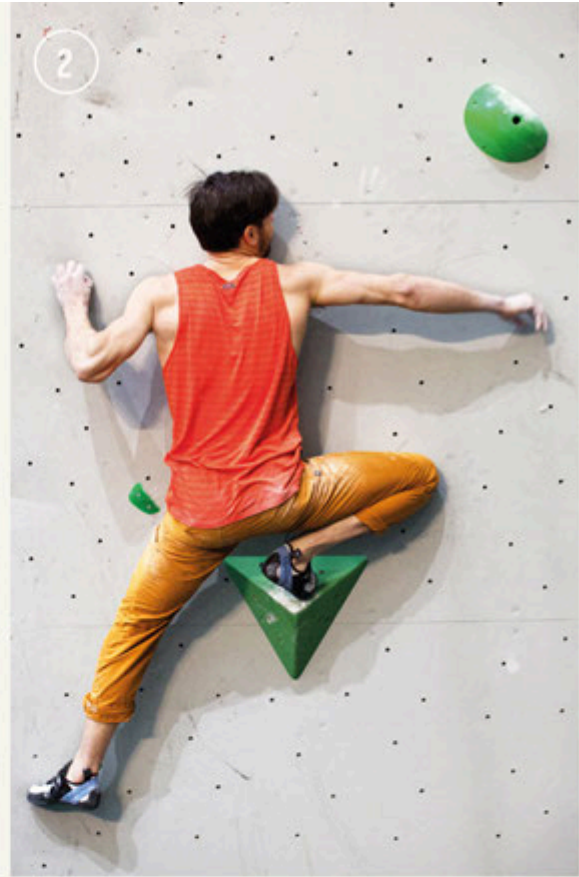
FRONTING

The most common climbing technique is what we call fronting. As the name implies, the chest should be facing the wall, knees pointing out to each side and the hips close to the wall. When moving your right hand, you should first move your hips sideways to the right to maintain balance, and vice versa when moving the left hand. When fronting, it's usually advantageous to lower your body to shift as much weight as possible on to your feet and get a better angle in relation to the handholds. This requires flexible hips. If you try to lower your body without being sufficiently flexible in your hips your centre of gravity will be pushed out from the wall which in turn will increase the load on your fingers. If the handholds are poor you will in most cases lose grip and fall. If the holds are positive you'll have more room to move around, but with your centre of gravity too far out from the wall you will be forced to expend more energy than is necessary. Fronting is the best technique for understanding the importance of weight distribution and balance, where small shifts in body position can greatly influence your balance.

To improve at fronting it is a good idea to train on a vertical wall. You should be using poor handholds and relatively good footholds, and move from side to side to get a feel for when you are in and out of balance. Try to find the position which requires the least amount of energy to stay on the wall.

This technique can be used whether you choose to do a move statically or dynamically. Particularly when making harder and longer moves, doing the move dynamically can be wise in order to save energy. In this case try to use speed to transfer your centre of gravity above your hips and move easily to the next hold without using your arms too much. Fronting is often the technique of choice on vertical walls where the holds are bad and a static style is more appropriate.



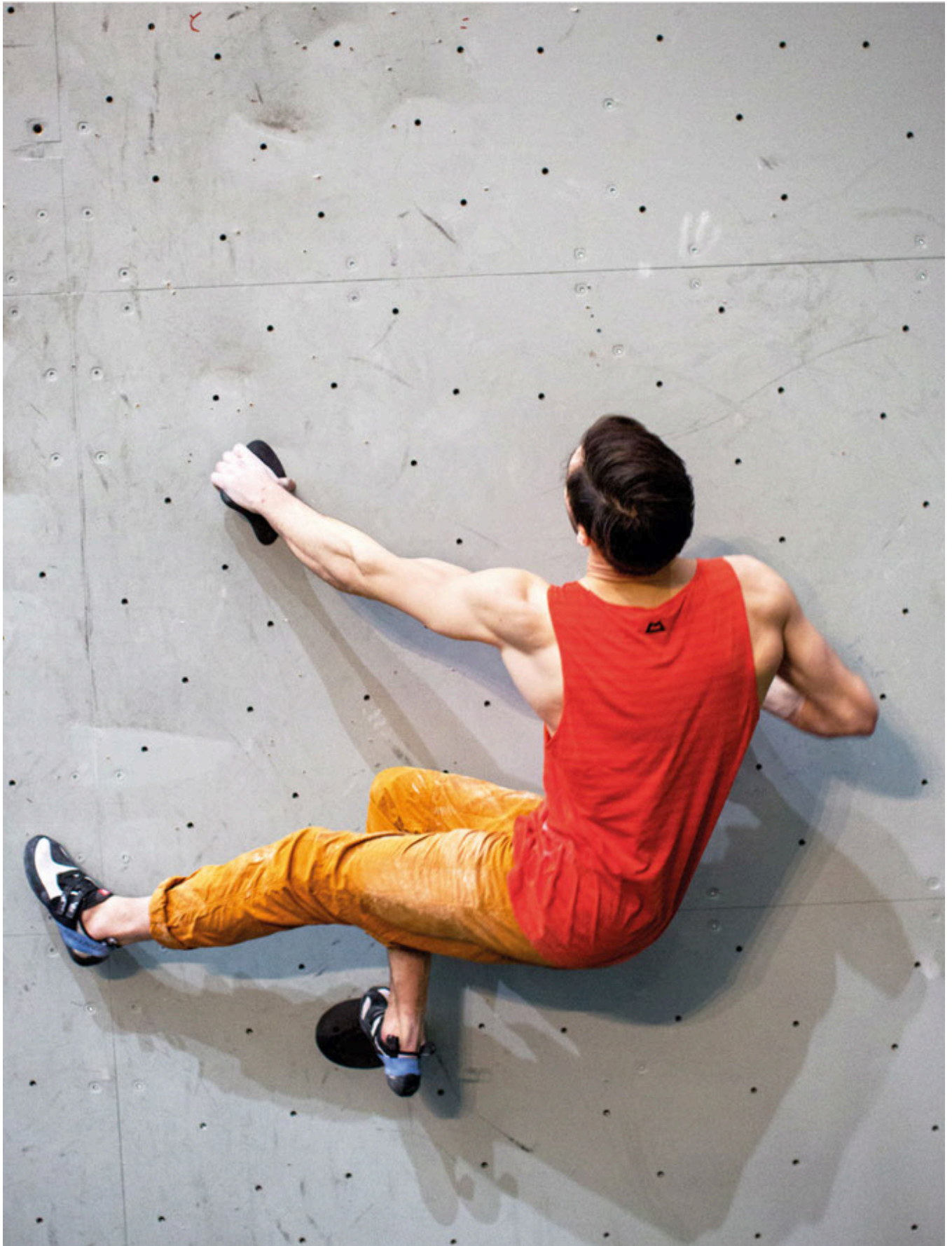


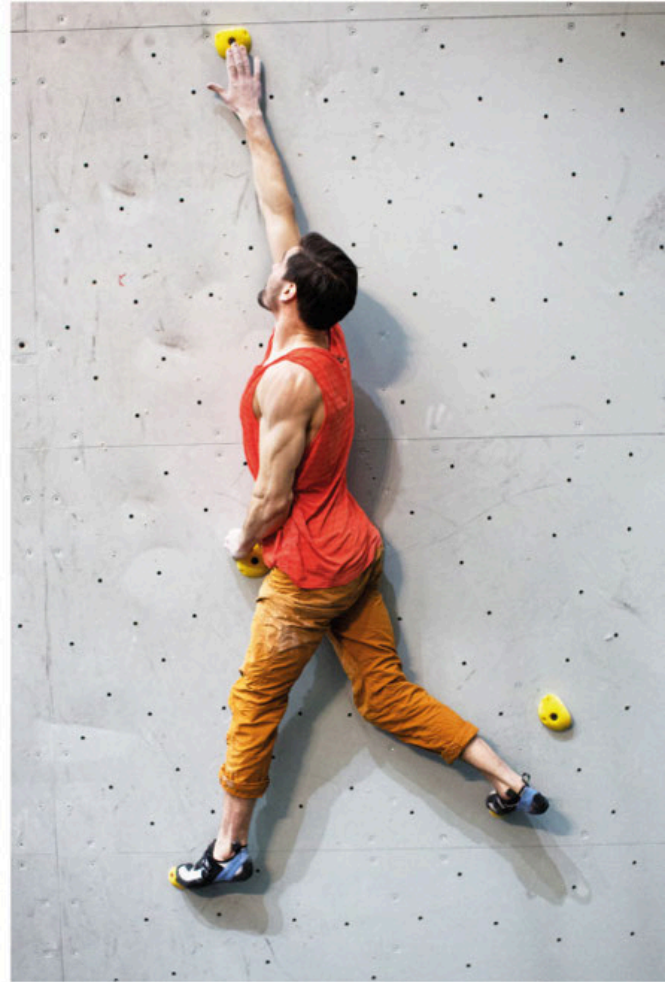
ROCKOVER

Rocking over is a specific type of fronting. This technique is useful when moving on to a high foothold and is performed by leaning your hips sideways over the foot, with your knee fully bent. By shifting your centre of gravity over your foot you'll reduce the load on your arms and keep your body up against the wall. This allows you to maintain grip on small and poor handholds. If you're not flexible enough to keep your body up against the wall it can help to place your heel on the foothold instead of your toes. This requires less hip flexibility, but does require a larger foothold. Rocking over is usually done statically and in control, when the holds you're moving from or to are very bad.

TWISTING

As an alternative to fronting, twisting shifts your hips sideways and positions your body to the side of the holds when doing a move. When moving your right hand, your right hip twists in towards the wall, and vice versa.





It's usually easier to reach further when twisting (right) than when fronting.

Twisting has three advantages:

- effective weight transfer on steep terrain
- ‘fuel efficient’ climbing
- increased reach.

This technique is frequently used when the wall is steeper than vertical. If we observe experienced climbers on steep terrain, we often see them twisting from side to side on every move. By twisting their hips into the wall, they increase their reach without having to pull themselves up with their arms, something which is of obvious advantage on steep walls. Twisting can also be used for vertical climbing, especially by those who aren't flexible enough to get their body close enough to the wall by fronting.

An extreme example of twisting is the drop knee. Drop knee is when we're twisting even further, sometimes even pushing the knee below the level of the foothold. This can be a useful technique when using a high foothold. The alternative would be to front, but using a drop knee requires less power to keep the body close to the wall and the centre of gravity centred on the area of support.



An example of a drop knee.



PHOTO: BJØRN HELGE RØNNING

Stian Christophersen flagging on Trøbbel (Font 8a+), Harbak, Norway.

FLAGGING

‘Flag like a nationalist’ is how Mats Peder Mosti describes the beta on his classic boulder problem Trøbbel at Harbak outside Trondheim in Norway.

Flagging is an alternative to twisting for finding balance through a move. Flagging is when you cross your free leg behind or in front of the leg you’re standing on. Which way you flag will depend on the height of the foothold. Flagging can be used on both vertical and overhanging climbing to create a larger area of support and to balance your centre of gravity.



In most cases where flagging is an option it will also be possible to match your feet and twist your hips in, but this might be difficult if the footholds are small, and it's usually more efficient to just cross your legs and flag. Flagging is a very useful technique on all wall angles as you can spend less energy doing moves.



PHOTO: BJØRN HELGE RØNNING

Hannah Midtbø winning the 2017 Nordic Bouldering Championship in Oslo.

The grip surface is the side of the hold you grip. The grip surface of a right-hand sidepull is on the right side. It's important to direct the force towards the grip surface, in this case towards the left.

SIDEPULLS, GASTONS AND UNDERCLINGS

In order to exert any force to the hold when climbing on sidepulls, it's important to find footholds which push or pull your centre of gravity to the opposite side of the grip surface. This ensures the direction of the force is directed on to the grip surface, allowing you to use the hold. It's helpful to try different foot placements to see how this affects your balance; it's often possible to find your balance by shifting your weight distribution, and you can switch between twisting and fronting when using sidepulls.

A gaston, or shoulder press, is when you use the opposite hand from that which you would normally use on a sidepull. This flips the script when it comes to body placement and balance. When using a gaston, your centre of gravity should be on the same side of the hold as the grip surface. This allows you to generate force and 'push' away from the hold. Your centre of gravity will then shift away from the handhold as you do the move. It is most common to front when using a gaston, but twisting can sometimes be the way to go. Using a gaston is physically demanding and requires strength in your shoulders and arms.



Joakim Louis Sæther has positioned his body so that he can use a left shoulder press to push himself to the right towards the next hold.



These three examples show the same move from an undercling but using a foothold at three different heights. In the first example the foothold is too low for the climber to be able to comfortably reach the next handhold, and in the second example the foothold is too high, pushing the climber's centre of gravity out from the wall. Example three shows the ideal foot placement.

An undercling is where the grip surface is underneath the hold. These holds can be used when there is a foothold below that allows force to be directed up into the handhold. In general, it's easier to use underclings if the footholds are higher, rather than lower, in relation to the handhold. This is because you can generate more power, and hence increase use of the undercling as your centre of gravity moves upwards. If your centre of gravity passes above the undercling, things start to get harder, as your weight is forced out from the wall. When making long moves which shift your centre of gravity above the undercling, it's advantageous to do these dynamically in order to minimise the time your body is in this physically demanding position.



LAYBACK

When climbing along an edge or a crack, or when using multiple sidepulls in sequence, it's impossible to ignore the technique known as laybacking. Laybacking is when you lean back and away from the holds. In the same way as when using a normal sidepull, your centre of gravity must be on the opposite side to the grip surface. The most common method is to twist and place your hips sideways against the wall, but laybacking can also be done by fronting. The position of the footholds will dictate the best way.

This balancing act is usually best performed statically. As your centre of gravity will always be near the perimeter of the area of support when laybacking, maintaining balance is a continuous challenge. In order to avoid swinging out like a barn door, we sometimes have to compensate by doing a move dynamically.



PHOTO: ØYVIND SALVESEN

Joakim Louis Sæther jamming on the Cellar Project (Font 7c+), Rogaland, Norway.

LOUIS' SCHOOL OF JAMMING

JAMMING IS WHEN you literally jam your hands, fingers or feet into cracks. This technique is most commonly used when climbing cracks outdoors. As we're not expert crack climbers, we've asked Joakim Louis Sæther to write some words about jamming:

Jamming hurts, but mastering this technique can be essential for sending certain routes and boulder problems. Jamming correctly is less painful, but it's still a good idea to tape your hands before you start. The following is a list of some different types of jams:

- finger jam
- hand jam
- fist jam
- stacking
- foot jam.



FINGER JAM

The best finger jams are also the scariest: this is when you can wedge your knuckles into a V-shape (a constriction) in the crack. Finger jams can be placed in many different ways. The thumb can be pointing either up or down, and you can use either one or more fingers in the crack. You should place your fingers into the crack with your elbow pointing out to the side, then twist your elbow down and along the crack. This way you can lock your fingers in place in a parallel crack without getting them completely stuck.



Louis'
SCHOOL OF JAMMING



HAND JAM

Pretend to be Master Yoda about to karate chop a rock into two pieces.

Place your flat hand into the crack, fold your thumb in towards your palm and flex your thumb muscle as hard as you can while pressing the tips of your fingers against the inside of the crack. This might hurt a little, but with the proper technique a good hand jam should feel like the best jug in the world! Twist your elbow down along the crack to secure the jam.



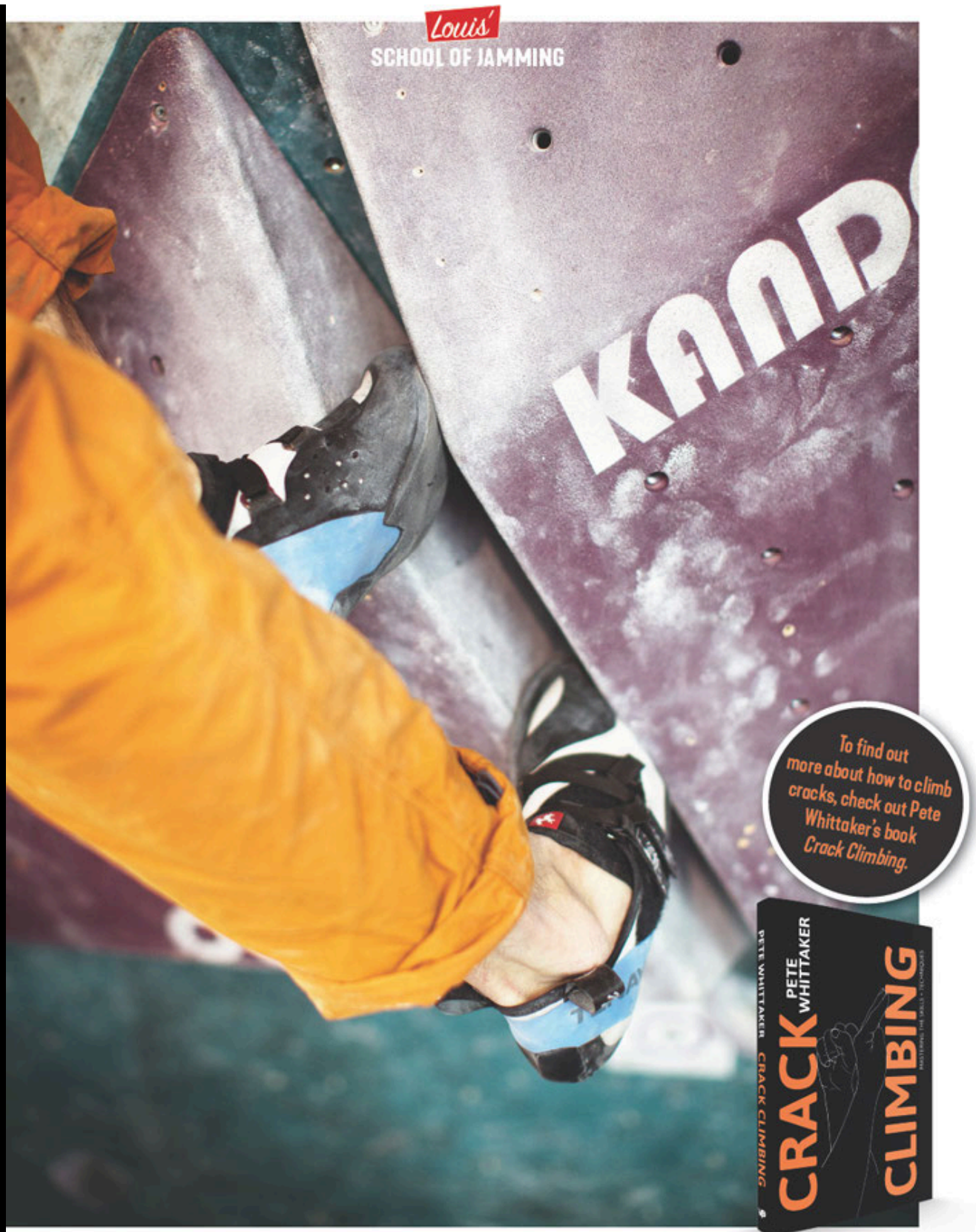
FIST JAM

Have you ever been so angry that you've clenched your fists, ready to throw punches?

Place your fist in an appropriately sized crack. Clench your fist and squeeze so that both sides of your hand expand and push against the sides of the crack. A fist jam is usually more painful than a hand jam. Again, twist your elbow in to secure the jam.

STACKING (COMBINATION JAMS)

If the crack is wider than a fist – usually called an offwidth crack – then you need to combine the different techniques. For example, you can stack two hands, or a hand and a fist, against one another. Which option you choose will depend on the size of the crack.



FOOT JAM

Easy, but painful! The most common way is to place your foot in the crack sideways with the pinkie toe facing down, and then twist your knee back up along the crack.



The climber lowers herself below the sloper to improve the angle and maximise friction.



She has to shift her centre of gravity out from the wall slightly in order to generate speed towards the next hold. It's important to move out of this position quickly.

CLIMBING ON SLOPERS

Many climbers find climbing on slopers to be the most challenging type of climbing. This is because it requires a great deal of skill when it comes to balance and weight transfer, and often in combination with dynamic moves. There are two factors that are especially important when climbing on slopers:

- The centre of gravity should be kept low and close to the wall.
- Movements should start from below and be dynamic.

Slopers have a negative angle and the force required to execute a move has to be generated from a position where the grip is at its best. This position is below the hold and close to the wall. If we try to pull up statically on a sloper, our centre of gravity will move out from the wall, resulting in a decrease in friction between hand and hold, with falling off being the consequence.



Jacky Godoffe writes in his book about route setting that they never dared to use deadpoint-style moves in lead competitions because the precision required to stick the move would be too high, and the chance of too many climbers falling on the same move would be too high. They only set these kinds of moves in bouldering competitions, where climbers could try the routes multiple times.



PHOTO: BÅRD LIE HENRIKSEN

The sit-start to Oppvarmingseggen (Font 6c+) at Harbak, Norway, is a classic compression problem on slopers.

The optimal way to move between slopers is to use the principle of the deadpoint. As we discussed in the section about dynamics, deadpointing is when you're able to stop at the apex of the move, when your body is 'weightless' and your centre of gravity is close to the wall. When done right, the move is executed from below, where the position is best, and uses as little energy as possible to reach the next hold. Stopping the motion at the deadpoint also minimises the strength required to hang on to the next hold.

As also described earlier, the force created between the points of contact on the wall is what we call tension, and this is central to climbing on slopers. By using slopers with the grip surface in opposing directions we can squeeze our hands together and thereby create more force and increase friction, which is essential to climbing on holds with a negative angle. This is known as compression climbing, and we often hear about 'compression boulder problems', or 'compression moves' when this is the defining attribute of a route or boulder problem. Compression climbing requires a strong upper body, as we use muscles in the upper arms, chest and shoulders to compress between the holds.





STEMMING

Stemming is a technique usually used when climbing inside corners (also known as dihedrals) and it works by pushing your arms and legs against two opposing faces. The angle of the corner influences how much of your weight can be shifted on to your feet. In narrow corners, at angles smaller than 90 degrees, we can almost solely rely on the friction created between hands, shoes and the wall, without the need for positive holds. At angles greater than 90 degrees, the difficulty increases, as does the need for handholds and footholds. The further away from the corner you place your hands and feet when stemming, the better. With a wide stance, you can often rest without using your hands at all. This does require a certain level of hip flexibility.

Climbing corners is usually a static endeavour, where time is spent finding the right placements with enough friction for your hands and feet.

MANTELING

To mantel – or mantelshelf – means to push your body up above your arm(s). This is particularly useful in two scenarios:

- when topping out a boulder problem or wall
- when scaling large features on the wall, such as a ledge or volume.



You can mantel by using both hands to raise your body over a ledge or feature so that you can get your feet up on to the same hold.



Manteling using only one hand can be a useful technique on large features, allowing you to reach up high to the next hold.

Manteling can be done by only using your arms, or more commonly with your feet as well. When performing a mantel it is essential to have your centre of gravity centred on the area of support. When manteling on to your right hand it's common to place your left foot high and to the left, and vice versa. Good hip flexibility is helpful as it will help you keep your body close to the wall. If you lose balance, there are no positive holds to help keep your body close to the wall. Manteling is usually done statically, but it can sometimes be helpful to start with a bit of speed to help you 'stand' with your arms.



Remember to angle your knee outwards to generate force inwards.
